White Wine Quality Prediction

Random Forest Classification and KMeans Clustering Analysis

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Index: 311197 | Group 2

Exploratory Data Analysis

1. Dataset

The dataset contains **3,934 observations** and **12 variables** describing white wines, including fixed acidity, residual sugar, total sulfur dioxide, sulphates, and alcohol. The target variable is **quality** - an expert rating on a scale from 1 to 7. Variables 1-11 are objective physicochemical measurements, while variable 12 (quality) is a subjective expert assessment.

The data has no missing values. There are **645 duplicates**, but for this type of data this is acceptable - different wines can produce identical chemical measurements, so duplicates were retained.

Feature distributions

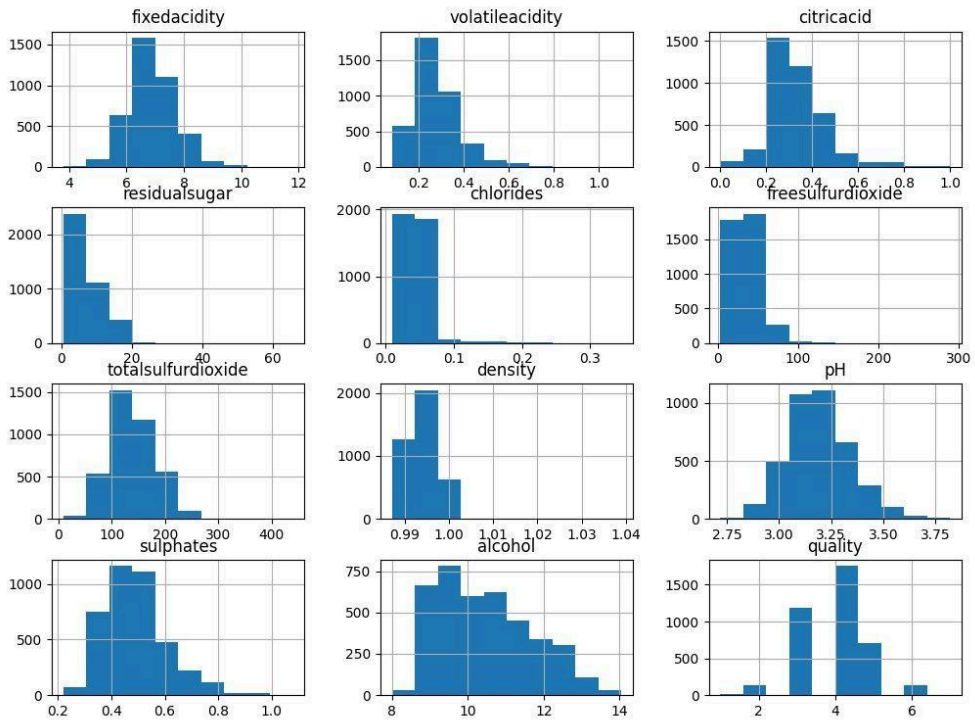

Figure 1. Histograms of all features in the dataset.

Most features show a right-skewed distribution. Typical ranges: fixed acidity 6-8, volatile acidity 0.3-0.4, citric acid 0.2-0.4, residual sugar 0-40, chlorides 0-0.1, free sulfur dioxide 0-80, total sulfur dioxide 100-130, density 0.99-1.0, pH 3.00-3.35, sulphates 0.4-0.6, alcohol 9-11. Quality (target) is most often rated 3, 4, or 5; values 1, 2, 6, and 7 are rare.

2. Correlation Analysis

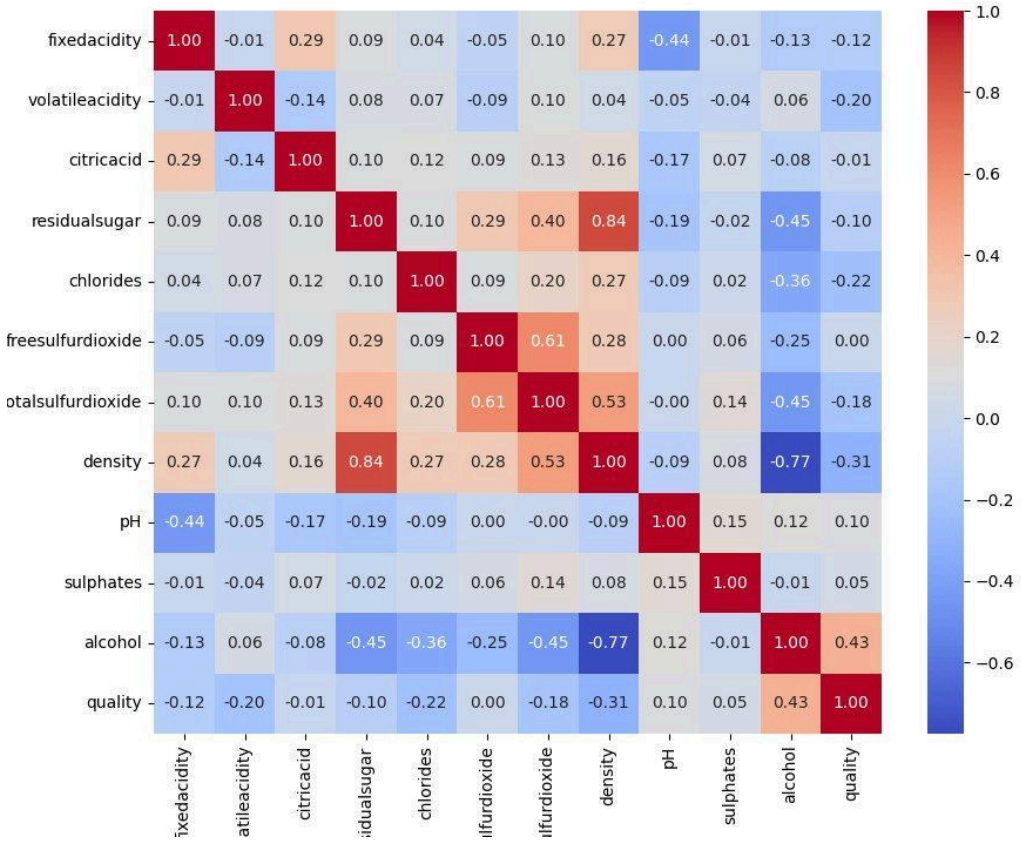

Figure 2. Correlation heatmap. The bottom row shows correlation with the target (quality).

Strongest correlations with quality

Feature	Correlation	Strength
alcohol	**+0.43**	**Strong positive**
density	**-0.31**	**Strong negative**
chlorides	-0.22	Moderate negative
volatileacidity	-0.20	Moderate negative
totalsulfurdioxide	-0.18	Moderate negative
fixedacidity	-0.12	Weak negative
pH	+0.10	Weak positive
residualsugar	-0.10	Weak negative
sulphates	+0.05	Negligible
freesulfurdioxide	0.00	None
citricacid	-0.01	None

High wine quality is most strongly associated with **high alcohol content** and **low density**. Note also a strong negative correlation between density and alcohol (-0.77), and a strong positive correlation between density and residual sugar (+0.84) - these features are not independent.

3. Classification Model - Random Forest

Configuration

Parameter	Value
Number of trees	300
Random state	311197 (index number)
n_jobs	-1 (all CPU cores)
Train/test split	70/30, stratified by quality
Features	All 11 physicochemical variables

Test set results

Metric	Value
Accuracy (exact)	0.6579
Accuracy with +/- 1 tolerance	0.9670
MAE	0.3785

Confusion matrix

True \ Pred	1	2	3	4	5	6	7
1	0	0	3	2	0	0	0
2	0	8	24	9	1	0	0
3	0	2	227	122	5	0	0
4	0	0	61	423	41	1	0
5	0	0	2	96	109	4	0
6	0	0	0	14	15	10	0
7	0	0	0	1	1	0	0

The model is most accurate for the dominant classes 3 and 4. Most prediction errors differ from the true value by only 1, which explains the high tolerance accuracy (96.7%). MAE of 0.38 confirms that on average, predictions are very close to the actual quality.

4. Clustering Model - KMeans

Features were standardized using StandardScaler. The optimal number of clusters was determined using the elbow method (inertia/SSE) and silhouette score for k from 2 to 12.

Elbow method

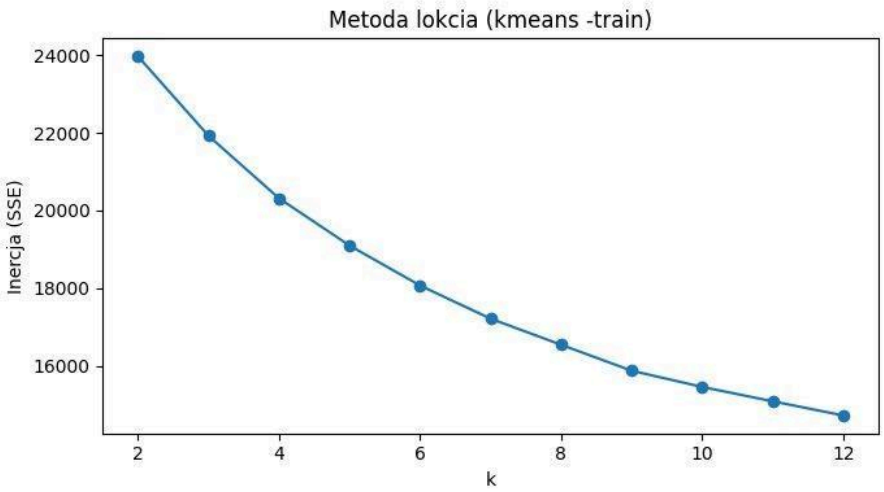

Figure 3. Elbow method - SSE decreases steadily as k grows.

Typical decreasing pattern with no sharp bend in the middle of the plot, making the elbow method alone inconclusive for choosing the optimal k.

Silhouette score

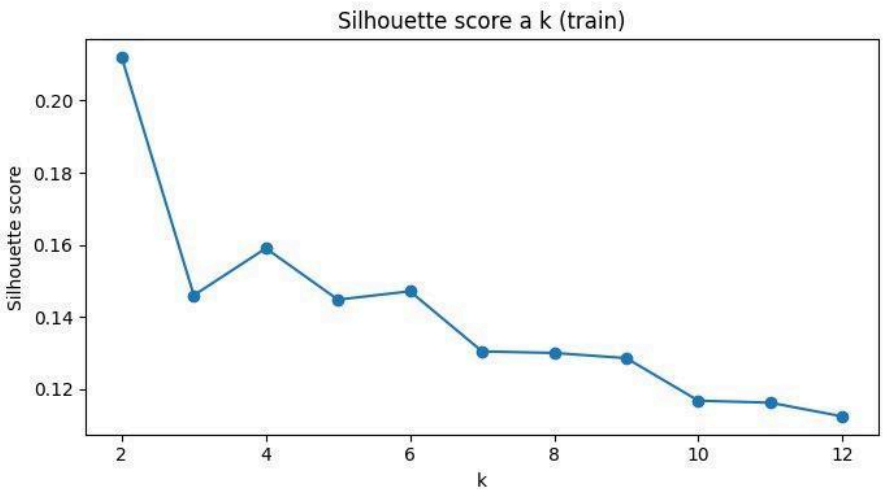

Figure 4. Silhouette score per k - highest at k = 2.

The silhouette score is highest for **k = 2** (~0.21), with a clear drop for k = 3 and gradual decline afterwards. **Best k = 2** was chosen for the final model.

Silhouette samples

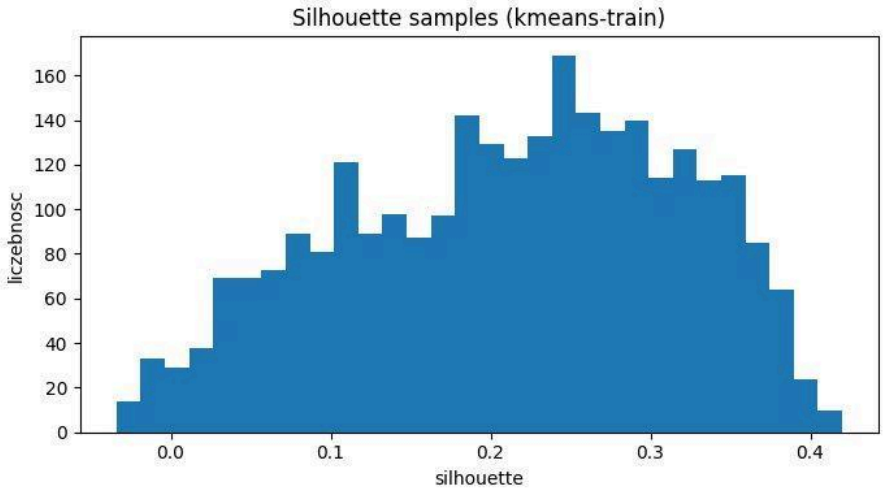

Figure 5. Distribution of silhouette samples for k = 2.

Most samples fall in the 0.2-0.3 range (count 120-140 per bin), confirming moderate cluster separation. Very few samples have negative silhouette values, meaning almost no points are misassigned.

5. Cluster Profiles

Cluster centroids (standardized features)

Feature	Cluster 0	Cluster 1
density	+0.93	-0.93
residualsugar	+0.85	-0.85
totalsulfurdioxide	+0.76	-0.76
freesulfurdioxide	+0.57	-0.57
chlorides	+0.39	-0.39
citricacid	+0.23	-0.23
fixedacidity	+0.19	-0.19
sulphates	+0.08	-0.08
volatileacidity	+0.07	-0.07
pH	-0.21	+0.21
alcohol	-0.77	+0.77

Cluster 0 - heavier wines: higher density, sugar, sulfur dioxide, and chlorides. Lower alcohol content. These wines tend to receive lower quality ratings.

Cluster 1 - lighter wines: lower density, sugar, and sulfur dioxide. Higher alcohol and pH. These wines tend to receive higher quality ratings.

Mean quality per cluster

Cluster	Train	Test
Cluster 0	3.57	3.63
Cluster 1	4.06	4.02

The clustering scheme generalizes well from train to test - mean quality per cluster is nearly identical in both sets. The split between clusters does correspond to a real difference in average quality (~0.4 points), confirming that the chemical features that separate the clusters are also linked to quality assessment.

6. Conclusions

Random Forest classification

- Predicts wine quality reasonably well: ~66% exact accuracy, ~97% accuracy within +/- 1.
- MAE of 0.38 means predictions are on average less than half a quality level off.
- Best for the dominant quality classes (3 and 4); rare classes (1, 2, 7) are harder to predict due to data imbalance.

KMeans clustering

- Optimal k = 2 selected based on silhouette score.
- Two clusters separate wines mainly by alcohol content, density, and sugar.
- Cluster 1 (lighter, more alcoholic wines) has higher average quality than Cluster 0.
- Cluster assignments generalize well to the test set.

Limitations and future work

- **Class imbalance** - quality classes 1, 2, 6, 7 are underrepresented. Resampling techniques (SMOTE) or class weighting could improve performance on these classes.
- **Hyperparameter tuning** - GridSearchCV could optimize Random Forest parameters (n_estimators, max_depth, min_samples_leaf).
- **Regression approach** - since quality is ordinal, treating it as a regression problem might better leverage the natural ordering of classes.
- **Multicollinearity** - density correlates strongly with both alcohol (-0.77) and residual sugar (+0.84). Feature selection or PCA could reduce redundancy.